

CV of the researcher

Personal Information

Family name, First name : Renaud, Ophélie

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Research Areas

- High-Performance Computing
- Rapid Prototyping
- Dataflow Programming
- Resource Allocation Optimization
- Simulation
- Radio-interferometric imaging pipelines

Dr. Ophélie Renaud's research addresses the challenges of high-performance computing (HPC) systems by optimizing resource usage, improving software productivity, and advancing the co-design of architectures and applications. She presents methods for optimizing resource allocation in multi-core processors, distributing resources across heterogeneous processors, and identifying optimal topologies for HPC applications. These contributions have been implemented in the rapid prototyping tool PREESM and are part of the development of the SimSDP co-simulator for radio astronomy. Her first post-doctoral research focuses on the automatic exploration of radio-interferometric imaging algorithms and the optimization of HPC architectures.

Education

-  **PhD** 2021–2024 Univ Rennes, INSA Rennes, CNRS, IETR
Specialization : *Signal, Image, Vision*
Thesis Title : *Granularity Optimization Based on a Model for High-Performance Computing Systems in Astronomy*
Supervisors : *Jean-François Nezan (Supervisor) and Karol Desnos (co-supervisor)*
-  **Engineering Degree** 2018–2021 INSA Rennes
Specialization : *Electronics – Design and Development of Innovative Technologies in Work-Study Program*
Project : *Digital Vision in Robotic Welding*
Supervisors : *Florent Hodiesne (Work-study Mentor) and Philippe Menke (Academic Tutor)*
-  **DUT (University Technology Diploma)** 2016–2018 University of Rennes
Specialization : *Electrical Engineering and Industrial Computing*
-  **High School Diploma** 2016 F.R. Chateaubriand High School, Combours
Series : *Scientific*
Section : *European Spanish*
Area : *Physics*
Distinction : *Swimming*
Mention : *With Honors*

Employment

-  **Postdoctoral Researcher** *October 2024–March 2026* IRISA, ENS Rennes, SATIE Paris-Saclay, France
Modeling dataflow applications for radioastronomical observatories. Optimization of radio imaging algorithms using AAM (Algorithm-Architecture Matching) optimization techniques, and participation in the NenuFar project to enhance data processing performance.
-  **Visiting PhD Researcher** *April 2024* Swinburne University, Melbourne, Australia
Collaboration with astronomers at Swinburne University as part of the RISE International Network for Solutions Technologies and Applications of Real-time Systems (Rising STARS). Dataflow modeling of a correlator as a case study for an automatic resource allocation method on heterogeneous CPU-GPU systems.
-  **Visiting PhD Researcher** *July–August 2023* CSIRO, Sydney, Australia
Collaboration with astronomers at CSIRO as part of the RISE International Network for Solutions Technologies and Applications of Real-time Systems (Rising STARS). Dataflow modeling of a radio frequency interference (RFI) filter as a case study for an automatic resource allocation method on HPC systems.
-  **PhD Researcher** *2021–2024* Univ Rennes, INSA Rennes, CNRS, IETR, France
Research on resource optimization for high-performance computing systems in astronomy. Development of methods for efficient data stream management in the SKA (Square Kilometre Array) project, integrating parallelization models and workload distribution.
-  **Founder & Designer** *2019–2023* AQUA-FIN, France
Design of swimming equipment with fins for competition using 3D printing and development of electronic systems.
-  **Application Engineer Apprentice** *July–August 2020* YASKAWA, Slovenia
Development of programs for industrial automation and human-machine interfaces (HMIs). Contribution to the implementation of an automated system to improve production performance in industrial robotics.
-  **Application Engineer Apprentice** *2018–2021* YASKAWA, France
Development of software and automation solutions for industrial robotic systems. Programming of PLCs and HMIs, and implementation of software for production and automated equipment management.
-  **Robotics Technician Intern** *April–June 2018* YASKAWA, France
Participation in the development of human-machine interfaces (HMIs) for managing industrial robots. Assisting teams in integrating new technologies into existing systems.
-  **Lifeguard** *Summer 2017 and 2018* SNSM, France
Responsible for beach and swimming surveillance. Providing emergency first aid and managing risky situations. Coordinating rescue team interventions to ensure the safety of individuals.
-  **Electrical Technician Intern** *March 2013* COBAC, France
Assisting with electrical installations and customer relations.

Educational Activities

Colloquium / Seminar / Lecture

-  **SORS seminar** *July 2025* BSC-CNS, Barcelona
Toward Automated Static-Dynamic Co-Design for Real-Time Radio Astronomy on Heterogeneous HPC.
[Presentation link](#)
-  **ICR seminar** *January 2025* ICR working group, France
SimSDP : Proof of Concept for Radio Astronomy Imaging on High-Performance Architectures.
-  **MAGELLAN seminar** *November 2024* IRISA, France
Optimizing Radio Astronomy Imaging Algorithms on HPC Systems with SimSDP and Fine-Grained Descriptions.
-  **VAADER seminar** *November 2024* INSA, IETR, France
Automated Deployment of Radio Astronomy Pipeline on CPU-GPU Processing Systems : DiFX as a Case Study. [Presentation link](#)
-  **ATNF, CSIRO colloquium** *May 2024* Swinburne University and CSIRO, Australia
Design and programming of heterogeneous and high-performance computing systems in astronomy. [Presentation link](#)
-  **VAADER reading group** *November 2023* INSA, IETR, France
Energy-efficient and High-throughput CNN Inference on Embedded CPUs-GPUs MPSoCs. [Presentation link](#)
-  **Green seminar** *October 2023* INSA, IETR, France
Design and programming of low power and high performance computing systems in astronomy.
-  **VAADER seminar** *October 2023* INSA, IETR, France
SCAPE : HW-Aware Clustering of Dataflow Actors for Tunable Scheduling Complexity. [Presentation link](#)
-  **ATNF, CSIRO colloquium** *August 2023* CSIRO, Australia
Design and programming of heterogeneous and high-performance computing systems in astronomy. [Presentation link](#)
-  **VAADER reading group** *March 2023* INSA, IETR, France
Automatically Scheduling Halide Image Processing Pipelines. [Presentation link](#)

Contest

-  **Hackathons ECLAT** *february-june-october 2025* Paris Observatory, France
Distributed measurementSet simulation.
-  **3MT contest** *August 2024* 32nd European Signal Processing Conference (EUSIPCO), France
[Presentation link](#)
-  **My PhD in 180 Seconds** *May 2024* National competition, departmental finalist, France
[Presentation link](#)

Supervision

-  **Supervisor for Master's Thesis Project** *February–August 2024* INSA, IETR, Swinburne University
Mentored a master's student internship on automatic resource allocation and optimized code for CPU-GPU systems using the DiFX correlator in radio astronomy pipelines, in collaboration with Australian astronomers, as part of the Rising STARS project.

Teaching

-  **Communication Bus Practical Sessions** *November–December 2023* INSA Rennes
BAC+4 : *20 hours*
This module covers the CAN protocol, where students used CubeMX and Keil to configure communication, handle frames, and manage interruptions.
-  **Parallel Programming on Embedded MPSoCs Practical Sessions** *October 2023* INSA Rennes
BAC+5 : *12 hours*
This module covers parallel programming for embedded MPSoCs using PThreads, OpenMP, and PREESM, with a project on parallelism and performance optimization.
-  **C Language Project Supervision** *2021 - 2022* INSA Rennes
BAC+4 : *16 hours*
This module covers C projects, focusing on design, coding, testing, best practices, Git, and documentation to build programming and teamwork skills.
-  **Mathematics Tutoring** *2020 - 2021*
9-10th Grade : *14 hours*
Tutored 9th and 10th-grade students in mathematics, providing concept explanations, exercises, and personalized support.

Online Tutorials

-  **DFT/FFT/G2G tutorial for beginner** *2025* DARK ERA
Basics of spectral transforms for NenuFAR imaging. [Tutorial link](#)
-  **DDFacet tutorial for beginner** *2025* Labcom ECLAT
Advanced LOFAR imaging with DDFacet. [Tutorial link](#)
-  **CPU-GPU Design Space Exploration** *2024* PREESM website
Optimization techniques for GPU-accelerated systems. [Tutorial link](#)
-  **SimSDP : Multinode Design Space Exploration** *2024* PREESM website
Multinode rapid prototyping for HPC architectures. [Tutorial link](#)
-  **SCAPE Clustering** *2024* PREESM website
Clustering for efficient multicore dataflow execution. [Tutorial link](#)

Online Application

-  **MeasurementSet simulator** *2024*
Synthetic interferometric data generation for distributed benchmark. [Application link](#)
-  **Distributed Generic Imaging Pipeline** *2024*
Dataflow-based radio-interferometric imaging pipeline for distributed systems. [Application link](#)
-  **MAD-based RFI filter** *2023*
Dataflow-based MeerKAT radio frequency interference pipeline. [Application link](#)

Research Grants

- Marie Skłodowska-Curie Fellowship under the European Union's Horizon 2020 research and innovation program, grant agreement No 873120.
- DARK-ERA (ANR-20-CE46-0001-01), funded by the French National Research Agency (ANR).

Publications

- **Ophélie Renaud**, Sunrise Wand and Nicolas Gac, *Exploring Algorithmic and Distributed HPC Strategies for Radio Astronomy Using SimSDP*, under review at Astronomy & Computing.
- **Ophélie Renaud**, Adrien Gougeon, Karol Desnos, Chris Phillips, John Tuthill, Martin Quinson, and Jean-François Nezan, *SimSDP : Automatic Workload-Balancing on Multi-Node & Multi-Core HPC Architectures based on dataflow models*, under review at IEEE Workshop on Signal Processing Systems (SiPS).

This article represents one of my most significant contributions because it introduces SimSDP, the first co-simulation framework that couples dataflow-based static modeling with dynamic execution-time simulation on distributed HPC systems. The method automatically balances workloads across both inter-node and intra-node levels, using iterative graph restructuring informed by SimGrid feedback. This approach addresses a critical challenge for emerging radio astronomy pipelines such as SKA imaging : the need to efficiently scale complex workflows to heterogeneous and massively parallel architectures.

The paper provides three major contributions :

(1) a rapid prototyping methodology enabling algorithm–architecture co-design before deployment ; (2) a model-based granularity optimization mechanism leveraging SCAPE to tune performance and reduce scheduling overhead ; (3) experimental validation on representative radio astronomy kernels, demonstrating up to 30–50% reduction in execution imbalance and latency.

Because this work lies at the core of my PhD and forms the foundation of my ongoing research on algorithm–architecture matching for radio interferometry, I consider it a cornerstone of my scientific output.

- 2025 **Ophélie Renaud**, Nicolas Gac, François Orieux, and Cédric Viou, *SimSDP, a Rapid Prototyping tool for Radio Astronomy : From NenuFAR Experiments to SKAO-Scale Simulation*, at IEEE Workshop on Signal Processing Systems (SiPS). [Publication link](#)
- 2025 **Ophélie Renaud**, Sunrise wang, Nicolas Gac, *Extension du prototypage rapide pour les pipelines d'imagerie radioastronomique : Simulation avancée multi-nœud HPC*, at Groupe de Recherche et d'Etudes de Traitement du Signal et des Images (GRETSI) conference. [Publication link](#)
- 2025 Ewen Michel, **Ophélie Renaud**, Karol Desnos, Adam Deller, Chris Phillips, and Jean-François Nezan, *Automated Deployment of Radio Astronomy Pipeline on CPU-GPU Processing Systems : DiFX as a Case Study*, in the 34th Astronomical Data Analysis Software and Systems (ADASS) conference. [Publication link](#)

This publication is selected because it showcases the first complete dataflow decomposition of the DiFX correlator and demonstrates how detailed algorithmic modeling can enable automated resource allocation on heterogeneous CPU–GPU platforms. The work was performed in collaboration with Australian astronomers (CSIRO and Swinburne) and provides a general methodology for transforming existing radio astronomy software into an analyzable representation suitable for performance prediction, design-space exploration, and rapid prototyping.

The key contributions include :

(1) a fine-grained static dataflow model of the correlator, enabling accurate memory and communication characterization ; (2) a heterogeneous mapping strategy evaluating multiple CPU–GPU configurations ; (3) automatic performance estimation demonstrating significant computational and energy-efficiency gains compared to manual deployments.

This work is important because it validates the practical usefulness of my modeling and optimization framework on a real, operational radio astronomy instrument. It also contributes directly to improving the scalability and deployability of correlators for next-generation telescopes, aligning with ASTRON's long-term research priorities.

- 2024 **Ophélie Renaud**, *Model Based Granularity Optimization for High Performance Computing Systems in Astronomy*, Thesis. [Publication link](#)
- 2024 **Ophélie Renaud**, Hugo Miomandre, Karol Desnos, and Jean-François Nezan, *Automated Level-Based Clustering of Dataflow Actors for Controlled Scheduling Complexity*, in the Journal of Systems Architecture (JSA). [Publication link](#)
- 2024 **Ophélie Renaud**, Erwan Raffin, Karol Desnos, and Jean-François Nezan, *Multicore and Network Topology Codesign for Pareto-Optimal Multinode Architecture*, in the 32nd European Signal Processing Conference (EUSIPCO). [Publication link](#)

- 2023 **Ophélie Renaud**, Naouel Haggui, Karol Desnos, and Jean-François Nezan, *Automated Clustering and Pipelining of Dataflow Actors for Controlled Scheduling Complexity*, in the 31st European Signal Processing Conference (EUSIPCO). [Publication link](#)
- 2023 **Ophélie Renaud**, Dylan Gageot, Karol Desnos, and Jean-François Nezan, *SCAPE : Regroupement d'Acteurs flux de Données en Fonction de l'Architecture pour une Complexité d'Ordonnancement Contrôlée*, at Groupe de Recherche et d'Etudes de Traitement du Signal et des Images (GRETSI) conference. [Publication link](#)
- 2023 **Ophélie Renaud**, Dylan Gageot, Karol Desnos, and Jean-François Nezan, *SCAPE : HW-Aware Clustering of Dataflow Actors for Tunable Scheduling Complexity*, in the Design and Architecture for Signal and Image Processing (DASIP) conference. [Publication link](#)

Other information

Programming languages

Python, C, C++, JAVA, CUDA, bash, HTML.

Target architectures

Grid5000 cluster, Ruche mesocentre cluster, Nancep cluster, CPU, GPU, PYNQ-Z2, SLURM, OAR.

Languages

 French (Native)

 English
(B2 TOEIC)

 Spanish
(A2/B1 DELE)

 Portuguese (Brazil)
(A2/B1)

Leisure

Swimming, fin swimming, triathlon, marathon, swimrun, climbing, 3D printing, Aqua-Fin.